Date:10-10-2026

CO1-ASSESSMENT TOOL-1

APPLICATION BASED QUESTIONS

COURSE CODE:DSA0308

COURSE NAME:NATURAL LANGUAGE PROCESSING

Reg no:192324194

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Question 1 – Resume Information Extraction

- First, we store the resume information as a text string.
- We import Python's Regular Expression (re) module.
- We use a regular expression to identify the candidate's name from the resume.
- Another pattern is used to find the email address.
- We use a pattern starting with 6-9 followed by 9 digits to identify the 10-digit mobile number.
- We create a list of technical skills such as Python, Java, SQL, Machine Learning, and NLP.
- The program checks whether each skill is present in the resume using regular expressions.
- Then, we extract the number of years of experience.
- Finally, the extracted information is displayed as a structured candidate profile.
- The candidate is considered eligible if they have at least 2 years of experience and Python skill.

In simple words:

The system takes a resume as input, searches the text using regular expressions, extracts important candidate details, and automatically checks whether the candidate satisfies the minimum requirements.

Sample output:

----- CANDIDATE PROFILE -----

Name: Rahul Kumar

Email: rahul.kumar@gmail.com

Mobile: 9876543210

Skills: Python, Java, SQL, Machine Learning, NLP

Experience: 3 years

----- ELIGIBILITY -----

Candidate is ELIGIBLE

Question 2 – Product Search System

- First, we create a list containing different product names.
- We import the Python re module for regular expression operations.
- The user provides a keyword and search type.
- For an exact search, the program looks for a product that exactly matches the keyword.
- For a prefix search, it checks whether the product name starts with the given keyword.
- For a suffix search, it checks whether the product name ends with the keyword.
- For a partial search, it checks whether the keyword appears anywhere in the product name.
- re.IGNORECASE is used so that searches are case-insensitive.
- All matching product names are displayed.
- Finally, the program counts and displays the total number of matching products.

In simple words:

The system searches a product list using different types of keyword matching. Regular expressions make the search flexible, and the system displays both the matching products and their total count.

Sample input:

Search Type: prefix

Keyword: Python

Sample optput:

Matching Products:

Python Programming Book

Python Laptop

Python Notebook

Total Matches: 3

Question 3 – University Registration System

- First, the student enters their register number, email, course code, semester, and mobile number.
- The program uses regular expressions to check whether each input follows the required format.
- The register number is checked to ensure it contains the required number of digits.
- The institutional email is checked to ensure it belongs to the required institutional domain.
- The course code is checked for the correct combination of letters and numbers.
- The semester is checked to make sure it is within the valid range.
- The mobile number is checked for a valid 10-digit Indian mobile number.
- For every field, the program displays either Valid or Invalid.
- A status variable keeps track of whether any validation has failed.
- At the end, if all fields are valid, the system displays Registration Successful; otherwise, it displays Registration Failed.

In simple words:

The system verifies all student details before registration. Regular expressions are used to validate each field, and registration is successful only when all the information is valid.

Sample input:

Enter Register Number: 2023123456

Enter Institutional Email: student@saveetha.com

Enter Course Code: CSE101

Enter Semester: 5

Enter Mobile Number: 9876543210

Sample output:

----- VALIDATION RESULTS -----

Register Number: Valid

Institutional Email: Valid

Course Code: Valid

Semester: Valid

Mobile Number: Valid

----- REGISTRATION STATUS -----

Registration Successful